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# Photometric Properties of Lunar Terrains Derived from Hapke's Equation

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**The photometric parameters of B. W. Hapke's (1986, *Icarus* 67, 264–280) equation are fit to the lunar disk-integrated visual lightcurve and to disk-resolved data of R. W. Shorthill, J. M. Saari, F. E. Baird, and J. R. LeCompte (1969, *Photometric Properties of Selected Lunar Features,* NASA Contractor Report CR-1429) for dark, average, and bright lunar terrains. The lunar nearside geometric albedo and phase integral computed from the disk-integrated results are consistent with those of earlier investigators. The single scattering albedos of disk-resolved average and bright lunar terrains are systematically larger than that of lunar mare. Average particles in dark terrain regoliths are more backscattering than those in average and bright lunar terrains. The angular width of the opposition surge is greatest in dark terrains and is found to be best explained by modest differences in regolith particle-size distributions which might accompany the normal regolith maturation process. The total amplitude of the opposition surge for dark terrains is larger than for average and bright terrains. This result appears to be a consequence of the fact that in opaque particles, a larger fraction of singly scattered light at zero phase comes from first-surface reflection. The average subcentimeter macroscopic roughness of dark terrains is significantly lower than that of average and bright terrains. The relative magnitude of this difference is consistent with that obtained from radar measurements at decimeter scales.** © 1987 Academic Press, Inc.

## INTRODUCTION

The ability to derive physically and geologically interpretable information from photometric measurements of planetary surfaces has evolved dramatically in the past 2 decades. Major advances in planetary photometry have arisen from the derivation of new photometric equations from radiative transfer theory (Hapke 1963, 1966, 1981, 1984, 1986, Irvine 1966, Emslie and Aronson 1973, Goguen 1981, Lumme and Bowell 1981a), the recent acquisition of radiometrically calibrated, high-resolution digital images returned by planetary space probes, and the widespread availability of modern computers.

Hapke's (1981, 1984, 1986) bidirectional reflectance equations provide the most rigorous description available of photometric behavior in terms of physically meaningful parameters. In the most recent version (Hapke 1986), photometric behavior is characterized by six model parameters: The efficiency with which average particles scatter and absorb light is described by the average particle single scattering albedo, $w$. The opposition effect (Seeliger 1887, 1895, Bobrov 1962, Hapke 1963, 1966, 1986, Gehrels *et al.* 1964, Irvine 1966, Whitaker 1979, Goguen 1981, Lumme and Bowell 1981), a surge in brightness observed in particulate surfaces near zero phase, is characterized by two parameters. The first, $h$, relates the angular width of the opposition surge to the combined effects of rego-



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lith compaction, rate of compaction with depth, and particle-size distribution; the second $S(0)$, describes the contribution of light scattered from close to the surface of an average particle at zero phase. The particle phase function is modeled by a Legendre polynomial with first- and second-order coefficients $b$ and $c$, respectively. The last parameter, $\bar{\theta}$, is a mean topographic slope angle which provides a measure of macroscopic roughness at subresolution scales. For reference, these parameters are summarized in Table I.

Laboratory investigations (Hapke and Wells 1981, Johnson *et al.* 1983, Clark *et al.* 1987, Mustard and Pieters 1987) have verified that Hapke's (1981) equation can predict successfully the bidirectional spectral reflectance of powdered glasses and binary mixtures of minerals on the basis of fundamental particle properties. In planetary studies, various versions of Hapke's (1981, 1984, 1986) equations have been applied to photometric observations of Mercury (Hapke 1984, Veverka *et al.* 1987a), Jovian and Saturnian satellites (Buratti 1985, Helfenstein 1986, Simonelli and Veverka 1987), and Uranian satellites and rings (Smith *et al.* 1986, Hapke 1986, Herbst *et al.* 1987, Thomas *et al.* 1987, Veverka *et al.* 1987b).

TABLE I

SUMMARY OF PHOTOMETRIC PARAMETERS

| Symbol | Description |
|---|---|
| | Hapke's parameters |
| $w$ | Single scattering albedo which characterizes the efficiency of an average particle to scatter and absorb light |
| $h$ | Characterizes the width of opposition surge in terms of soil structure (porosity, particle-size distribution, and rate of compaction with depth) |
| $S(0)$ | Opposition surge amplitude term which characterizes the contribution of light scattered from near the front surface of individual particles at zero phase |
| $b$, $c$ | Coefficients in the Legendre polynomial model particle phase function, $P(\alpha) = 1 + b \cos \alpha + c(1.5 \cos^2\alpha - 0.5)$ |
| $\bar{\theta}$ | Average topographic slope angle of surface roughness at subresolution scale |
| | Other related quantities |
| $B_0$ | Total amplitude of the opposition surge $B_0 = S(0)/w\{1 + b + c\}$ |
| $g$ | Asymmetry factor of the particle phase function. $g = \langle \cos \theta \rangle = -b/3$, where $\theta = \pi - \alpha$ is the scattering angle |

Remarkably, Hapke's most recent equations have not been applied rigorously to lunar photometric data, perhaps because, despite the wealth of photographic data, radiometrically calibrated digital information is limited to a modest sampling of Mariner 10 images and Earth-based near-opposition images digitized and calibrated from carefully controlled photographic products (Wildey 1978). An extensive body of lunar disk-resolved (Opik 1924, Bennett 1938, Sharonov 1939, Markov 1950, Fedoretz 1952, Sytinskaya 1953, van Diggelen 1959, Fessenkov 1962, Gehrels *et al.* 1964, Wildey and Pohn 1964, Rennilson *et al.* 1968, Saari and Shorthill 1967, Shorthill *et al.* 1969) as well as disk-integrated (Russell 1916a,b, Rougier 1933, Shorthill *et al.* 1969, Lane and Irvine 1973, Shevchenko 1982) photometric data have been compiled from nondigital sources. Calibration of some of these data were adequate for application of early versions of Hapke's equations (Hapke 1963, 1966, Gehrels *et al.* 1964, Wilson 1968, 1971, Shorthill *et al.* 1969).

The Moon is a very special object. It is the only extraterrestrial body from which samples have been returned to Earth for laboratory analysis, and on which extensive manned and unmanned geological observations have been conducted. It is also the body for which photometric coverage of individual regions is most complete in terms of variations in photometric geometry (illumination angle ($i$), viewing angle ($e$), and phase angle ($\alpha$)). Therefore, the lunar photometric data set provides an important test of the usefulness of Hapke's equation in deriving physically and geologically meaningful parameters. The values derived for the Moon also provide a basis for the comparison with parameters derived from other planet and satellite surfaces.

The complexity of Hapke's equation and comparatively large number of model parameters makes it difficult to find unique

meaningful solutions. A reliable method of solving simultaneously for all six parameters must be developed and tested. Most past efforts to apply Hapke's equation have involved reducing the number of parameters in the solution by *assuming* reasonable values for some of these (Hapke and Wells 1981, Hapke 1986, Herbst *et al.* 1987), by substituting simpler expressions for components of the Hapke equation, such as the particle phase function (cf. Buratti 1985, Goguen 1985, Simonelli and Veverka 1987, Thomas *et al.* 1987, Veverka *et al.* 1987b), or by focusing the study on some aspect of photometric behavior for which some of the parameters can be ignored (Johnson *et al.* 1983, Clark *et al.* 1987, Mustard and Pieters 1987, Hapke 1986, Herbst *et al.* 1987).

The primary objective of the present investigation is to derive Hapke parameters for the lunar surface from both disk-integrated and disk-resolved photometric data and to examine the mutual consistency of values derived for three principal albedo classes of lunar terrains. Fundamental photometric properties such as geometric albedo and phase integral are evaluated and compared to the results of previous investigators. Finally, the geological significance of results is considered.

## II. APPROACH

We begin by deriving Hapke parameters for the lunar nearside from disk-integrated photometric phase curves. The results are used to facilitate the determination of parameters for individual classes of lunar terrains from disk-resolved data as described below.

### *Disk-Integrated Data*

Over the last century, disk-integrated measurements for the lunar nearside have been compiled by a number of investigators (Russell 1916a,b, Rougier 1933, Shorthill *et al.* 1969, Lane and Irvine 1973). Of these, the most recent photoelectric measurements of Lane and Irvine (1973) are considered to be the most accurate. Unfortunately, the data of Lane and Irvine did not extend to phase angles low enough to characterize the opposition effect fully. In order to constrain the opposition parameters adequately, it is necessary to include some earlier measurements, after scaling for consistency with the Lane and Irvine data. It should be noted that the data used were obtained at slightly different wavelengths. The data of Shorthill *et al.* and Rougier correspond to an effective wavelength of 445 nm, while those of Russell are near 550 nm. Lane and Irvine present data over the spectral range from 350 to 1200 nm and, according to these authors, the geometric albedos of the Moon in the $B$ and $V$ wavelengths of the $UBV$ system are 0.094 and 0.113, a difference of about 10%.

The measurements of Shorthill *et al.* (1969), Rougier (1933), and Russell (1916a,b) were normalized to those of Lane and Irvine (1973) so that the disk-integrated brightness at zero phase angle, extrapolated linearly on a magnitude scale over phase angle ranges from 20° to 90° corresponded to a geometric albedo of 0.113. The data were then combined and subdivided into phase angle bins as follows: data in the range of 0°–10° were divided into 2.5° bins, data in the range of 10°–30° into 5° bins, and data in the range of 30°–150° into 10° bins. Average values of disk-integrated brightness and phase angle were determined for each bin and used in subsequent calculations (Fig. 2).

### *Disk-Resolved Data*

The lunar disk-resolved data used in this investigation are a subset of the extensive measurements tabulated by Shorthill *et al.* (1969). These data cover phase angles in the range of $-124.5°$ to $+135.7°$, where negative and positive phase angles distinguish data obtained before and after full moon, respectively. Data nearest to opposition are at $\alpha = -2.3°$. Shorthill *et al.* divided terrains into three groups on the basis of normal reflectances, $r_n$. Thirty-two dark lunar terrains ($0.06 < r_n < 0.09$), twenty-

five average terrains ($0.10 < r_n < 0.12$), and thirty bright terrains ($0.13 < r_n < 0.16$) are listed in their catalog of measurements.

Ideally, photometric parameters should be derived independently for individual regions. Unfortunately, phase angle coverage obtainable from Earth depends strongly on the region's location on the nearside. Regions near limb can be observed over a wide range of phase angles, while regions near the center of disk are visible over a more restricted range of phase angles (<90°). Optimal fitting of certain Hapke parameters, such as particle phase function and macroscopic roughness, require data at phase angles both below and above 90° (cf. Helfenstein 1986, 1987). Comparatively few of the regions sampled satisfy this requirement, and none is visible at both high negative and high positive phase angles.

An acceptable compromise is to combine data for all examples of a particular terrain class. Before adopting this approach, an important problem must be addressed. It is likely that, even within the same terrain class, there may be significant regional differences in albedo. While all regions were observed at low phase angles, only a few were observed at high ones. The combined raw data would represent a different average albedo at low phase angles than at high phase angles. Effectively, an artificial phase angle dependence would be introduced by a simple combination of brightness measurements for all terrains within a particular albedo class.

To circumvent this problem, some method must be found for correcting the brightnesses of all terrains within a class to a consistent albedo at all phase angles. Adjusting the brightness of each terrain region so that all have the same reflectance near zero phase is inadequate, first because systematic regional differences in opposition surge may exist, and second because different terrains are sampled over different incidence and emission angles at any given phase angle. The second consideration poses a problem for almost any method of correcting the data. In the case of Io, Simonelli and Veverka (1987) were able to combine disk-resolved data for different examples of similar terrains by first determining Minnaert (1961) photometric parameters for each terrain class at each phase angle, and then correcting all brightnesses to that predicted for the center of disk. Their approach took advantage of changing viewing geometries provided by the Voyager I and II encounters which enabled them to observe identical areas of Io at different incidence and emission angles. It would be difficult to apply this technique to the present Earth-based data which, for any given phase angle, provide only one incidence and emission angle for each terrain example.

The average lunar Hapke parameters derived from the disk-integrated data provide the best way of correcting raw data to a consistent albedo. With these parameters, it is possible to ratio the brightness of each data point for a given region to the brightness predicted for average lunar material under identical illumination and viewing conditions. Shorthill *et al.* (1969) provided plots of this ratio as a function of phase angle for each lunar region. The data points in these plots define curves which superficially resemble one another for terrains which are photometrically most alike. Similar ratio plots were successfully used by Helfenstein (1986) to distinguish different photometric classes of terrains on Ganymede.

We now illustrate how these ratio plots were derived and applied to normalizing data in our study. Let $R_j = r_j/r_{\text{ave}}$, where $r_j$ is the observed brightness of a region and $r_{\text{ave}}$ is the brightness predicted for average lunar material under identical illumination and viewing conditions. For average lunar material, $R_j = 1$ over all phase angles. For a material which differs from the average only by albedo, ratios on a $R$ vs $\alpha$ plot would define a line parallel to $R_j = 1$, but displaced vertically by a factor approximately equal to $w_j/w_{\text{ave}}$, where $w_j$ and $w_{\text{ave}}$

are the single scattering albedos of the region and average Moon, respectively. The ratios will not define a horizontal line on a $R$ vs $\alpha$ plot if the terrain differs from average by some property other than albedo. Consider, for example, a material which is macroscopically smoother than average. Because at high phase angles, positive relief casts shadows toward the observer, average lunar material, being rougher, would cast more shadows than the smoother terrain; thus $R_j$ would increase with phase angle. In summary, the shapes of $R$ vs $\alpha$ curves can be used to identify differences in large-scale roughness, opposition effects, or particle phase functions. Vertical displacements of similarly shaped $R$ vs $\alpha$ curves provide a crude measure of differences in albedo for terrains which are otherwise photometrically similar.

Given two terrains which differ only in albedo and for which at least a few observations exist at overlapping phase angles, it is possible to find a correction factor, $m_{jk}$, which can be multiplied by the brightness ratios $R_j$, of terrain $j$, to make them coincide with corresponding brightness ratios $R_k$ of terrain $k$. In a least-squares sense

$$m_{jk} = \frac{\sum_{l=1}^{p} R_j(\alpha_l)R_k(\alpha_l)/\sigma_l^2}{\sum_{l=1}^{p} R_j^2(\alpha_l)/\sigma_l^2},$$

where $l$ indexes overlapping phase angles, $\alpha_l$, and $p$ is the total number of such phase angles. The value of $\sigma_l$ may be interpreted as the average amount of misfit which would result if only data point $l$ were used in computing $m_{jk}$. More precisely,

$$\sigma_l^2 = \frac{R_k(\alpha_l)^2}{p} \sum_{n=1}^{p} \left[\frac{R_k(\alpha_n)}{R_k(\alpha_l)} - \frac{R_j(\alpha_n)}{R_j(\alpha_l)}\right]^2.$$

The simultaneous application of this approach to a large number of regions is summarized in Fig. 1. Figure 1a shows an uncorrected plot of $R$ vs $\alpha$ for the combined 32 regions of dark terrain. Figure 1b results from performing two operations on these data. First, each terrain sample $j$ is registered to a common sample $k$. Second, to ensure that the corrected ratios represent the actual average albedo (rather than arbitrarily assigning the "average" to be the albedo of sample $k$), the registered data are then divided by the average of all correction factors, $\bar{m} = \Sigma_{j=1}^{q} m_{jk}/q$, where $q$ is the total number of regions in the class set.

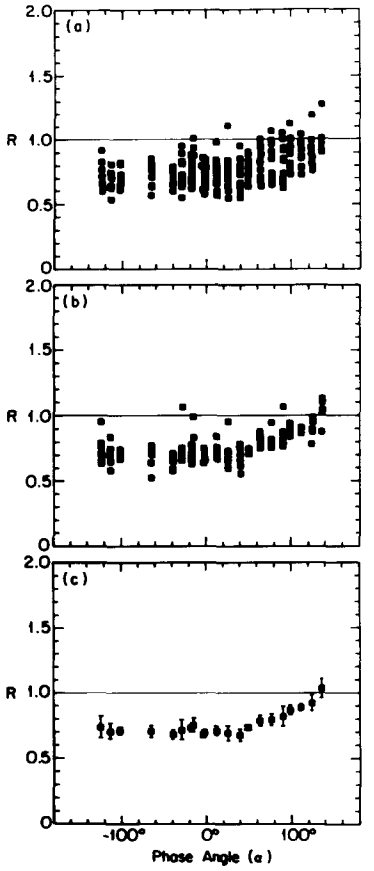



FIG. 1. Registration and averaging of disk-resolved data for dark terrains (see text for discussion). (a) $R$ vs $\alpha$ plots for all dark terrain samples combined. (b) $R$ vs $\alpha$ plots after all samples have been registered to a common albedo. (c) averaged data from (b) with one standard deviation error bars.

In Fig. 1c, the corrected ratios at each phase angle have been averaged so that there is a single point at each phase angle. Error bars are one standard deviation. In our calculation, the input parameters were obtained by the following process: a single

value of incidence and emission angle at each phase angle was taken as the mean of all incidence and emission angles within each phase angle bin. The brightness at each phase angle was computed from average lunar behavior multiplied by the corresponding ratio, $R_j$, at the corresponding phase angle.

## III. LEAST-SQUARES FITTING OF THE HAPKE EQUATION

Hapke's equation is highly nonlinear and cannot be solved in closed form. Iterative numerical approaches must be applied to determine best-fit parameters. The method used in this study is an improved version of that described in detail by Helfenstein (1986). The fundamental requirement is that values of parameters be found which minimize $\chi^2$, defined as

$$\chi^2 = \sum_n \frac{[r_n - r(i_n, e_n, \alpha_n, w, h, S(0), b, c, \bar{\theta})]^2}{\sigma_{r_n}^2},$$

where $r_n$ is the measured reflectance of data point $n$, $i_n$, $e_n$, $\alpha_n$, are the corresponding angles of incidence, emission, and phase, respectively, and $r(i_n, e_n, \alpha_n, w, h, S(0), b, c, \bar{\theta})$ is the reflectance predicted by Hapke's equation given the assumed model parameters. The uncertainty in the measured reflectance is represented by $\sigma_{r_n}$.

In an iterative algorithm, a plausible first-guess of model parameters is used to compute an initial value for $\chi^2$. Incremental corrections for the parameters are then found for which $\chi^2$ is reduced to a smaller value. The corrected values of model parameters are then used as estimates for a subsequent trial. The latter process is repeated until the smallest possible value of $\chi^2$ is found.

Many techniques have been devised for computing the actual incremental corrections. Mosher and Lorre (1981) applied a stepwise grid search of parameter space to solve an earlier, four-parameter version of Hapke's (1981) equation. Grid search algorithms require considerable computing time and are impractical for solution involving many parameters and large data sets. In this study, three passes through a coarse, stepwise grid search are employed at the beginning of the parameter search to minimize the adverse effects of a poor initial guess on a subsequent, more refined search procedure. The results of the coarse grid search are applied as initial values for the gradient-expansion algorithm of Marquardt (1963), as adapted by Bevington (1969). The salient feature of the gradient-expansion algorithm is its rapid rate of solution convergence for reasonable first-guesses of model parameters. At moderate parameter distances from the solution, the method follows the steepest gradient of the $\chi^2$ hypersurface (that is, the locus of $\chi^2$ as a function of all possible model parameter values). In the immediate vicinity of the solution, where the gradient shallows and a conventional gradient-follower would slow convergence, the algorithm automatically linearizes the equation to speed the convergence process.

## IV. RESULTS

### A. *Hapke Parameters from Disk-Integrated Data*

Values of the Hapke parameters were derived from the averaged disk-integrated data as follows. At each phase angle, a numerical integration of Hapke's bidirectional reflectance equation was used to determine the value of disk-integrated brightness used for computing $\chi^2$. Following the approach of Simonelli and Veverka (1987), an eight-point Gaussian quadrature was applied to the appropriate spherical geometry. The original computer subroutine used in their study was modified to eliminate the need for Hapke's (1984) macroscopic roughness correction tables. Initial-guess values for $w(0.25)$, $b(0.6)$, $c(0.4)$, and $\bar{\theta}$ (20°) were those reported in Hapke (1984) for the planet Mercury. The corresponding first-

guess value for $h$ (0.05) was that derived by Hapke (1986) from the lunar farside data of Whitaker (1979). $S(0)$ was initialized arbitrarily at unity.

Preliminary efforts to find simultaneously all six parameters resulted in physically implausible solutions. In an earlier study, Helfenstein (1986) reported such strong mathematical coupling of parameters in Hapke's equation that meaningful solutions could often be found only by sequentially solving parameters in the order of their sensitivity to the data distribution.

After many trials in which one parameter was held constant and the others allowed to vary, it appeared that the strongest coupling existed between $\bar{\theta}$ and the phase function parameter, $b$. In a new sequence of trials, a parameter search was performed in which $\bar{\theta}$ and $b$ were alternately held constant. The approach was iterative in the sense that the result of one trial was used as the initial guess for the subsequent trial. Trials were repeated in this way until all parameters converged to constant values. The top portion of Table II lists the numerical results. Figure 2 plots the data and model-integrated brightness as a function of phase angle. Note that the disk-integrated brightness has been normalized in such a way that the value shown at $\alpha = 0°$ is equivalent to the geometric albedo of the Moon.

Figure 2 shows that an excellent fit to the data was obtained. It is possible, however, for different combinations of Hapke parameters to predict similar disk-integrated phase curves. If our solution is valid, then

TABLE II

HAPKE PARAMETERS FOR LUNAR TERRAINS

| | $w$ | $h$ | $S(0)$ | $b$ | $c$ | $\bar{\theta}$ |
|---|---|---|---|---|---|---|
| Disk-integrated | 0.21 | 0.07 | 0.71 | 0.29 | 0.39 | 20.0 |
| Disk-resolved | | | | | | |
| Dark terrains | 0.12 | 0.12 | 0.21 | 0.41 | 0.10 | 8.1 |
| Average terrains | 0.25 | 0.06 | 0.78 | 0.33 | 0.37 | 20.6 |
| Bright terrains | 0.33 | 0.05 | 1.03 | 0.26 | 0.45 | 24.0 |

*Note.* Particle phase function $P(\cos\alpha) = 1 + b\cos\alpha + c(1.5\cos^2\alpha - 0.5)$.

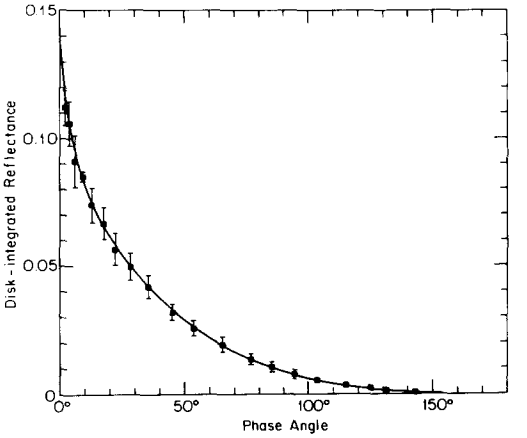



FIG. 2. Fit of Hapke's equation to disk-integrated lunar lightcurve (continuous curve). Brightness at zero phase is the geometric albedo. Plotted data points are normalized to be consistent with Lane and Irvine (1973) and averaged as described in text. Error bars are one standard deviation.

the parameters in the top row of Table II should be able to predict average systematic variations in brightness from limb to terminator at each phase angle. These variations can be characterized by the $B_m$ and $k$ parameters of the empirical Minnaert (1961) photometric function, $r = B_m(\cos i)^k(\cos e)^{k-1}$. Average lunar values of the Minnaert $B_m$ and $k$ were determined by J. Goguen (1980, private communication) from a random sampling of the Shorthill *et al.* (1969) disk-resolved data and are plotted as solid square boxes with error bars in Fig. 3. Values of $B_m$ and $k$ predicted from our solution (solid curves in Fig. 3) were evaluated by performing a nonlinear least-squares fit of the Minnaert equation to Hapke-computed brightnesses of 4000 to 9000 points distributed over the visible portion of a hypothetical lunar disk. The predicted values of $B_m$ and $k$ are in good agreement with those determined directly from the lunar photometric data.

Photometric properties of the Moon and Mercury at visual wavelengths are known to be remarkably similar (cf. Hapke 1977). Our values of $S(0)$, $b$, and $\bar{\theta}$ are nearly identical with those found by Veverka *et al.*

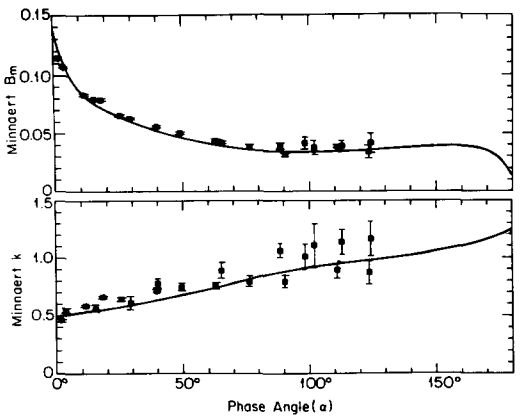



FIG. 3. Minnaert $B_m$ (top) and $k$ (bottom) as a function of phase angle. Solid line is predicted from a fit of Hapke's equation to disk-integrated data. Solid squares with error bars are averaged lunar values derived directly from the data of Shorthill *et al.* (1969) by J. Goguen (1980, private communication).

(1987a) for the planet Mercury. Our value of $w$ (0.21) is lower than Hapke's value for Mercury (0.25) based upon his earlier (1984) work, but is not markedly different from the new value (0.20) reported by Veverka *et al.* (1987a) derived using Hapke's (1986) equation. Our value of $h$ (0.07) falls in between the Veverka *et al.* value for Mercury (0.08) and the 0.05 value derived from lunar farside data by Hapke (1986). These modest numerical differences in $h$ may not be significant statistically.

The particle phase parameter $b$ can be expressed in terms of the well-known asymmetry factor and provides a further basis for comparison to earlier lunar results. The asymmetry factor, $g = \langle \cos \theta \rangle = -b/3$, where $\theta = \pi - \alpha$ is the scattering angle. Negative values of $g$ represent particles which are more strongly backward than forward scattering, and conversely, positive values identify particles which are dominantly forward scattering. Our equivalent value of $g = -0.1$ is identical to the earlier result of Lumme and Irvine (1982) for the Moon, derived with the Lumme–Bowell (1981a) photometric function.

Fundamental photometric parameters, such as geometric albedo, $p_v$, phase integral, $q_v$, and spherical albedo, $p_v q_v$ can be computed from our derived Hapke parameters and results compared with earlier determinations (Table III). Considering only those determinations in Table III obtained exclusively from nearside photometric data, we find that our calculated geometric albedo is intermediate between those of Harris (1961) and Lane and Irvine (1973), who ignored the opposition effect, and Lumme and Irvine (1982), who included it. Our estimated phase integral is most consistent with that determined by Lumme and Irvine. Both of these recent phase integrals are significantly lower than earlier estimates; both are derived from fits of model photometric equations to disk-integrated data. The more complete description of photometric behavior at very low and very high phase angles provided by these equations may explain why these two studies find mutually consistent values of $q_v$ which differ from previous estimates. Our value for the product of $p_v$ and $q_v$ (the spherical albedo) is somewhat below the values estimated by earlier investigators. Lane and Irvine (1973) found that, for the Moon, $A_{\text{Bond}}/p_v q_v = 0.123/0.069 = 1.78$, where $A_{\text{Bond}}$ is the radiometric Bond albedo. Using this scaling, our value of $p_v q_v$ corresponds to a Bond albedo of 0.11.

### *B. Hapke Parameters from Disk-Resolved Data*

Disk-resolved data for the three principal albedo groups (dark, average, and bright) were registered and averaged as discussed

TABLE III

LUNAR GEOMETRIC ALBEDO, PHASE INTEGRAL, AND SPHERICAL ALBEDO

| $p_v$ | $q_v$ | $p_v q_v$ | Source |
|---|---|---|---|
| 0.115 | 0.585 | 0.067 | Harris (1961) |
| 0.113 | 0.611 | 0.069 | Lane and Irvine (1973) |
| 0.152 | 0.476 | 0.072 | Lumme and Irvine (1982) |
| 0.147 | 0.509 | 0.075 | Shevchenko (1982)[a] |
| 0.136 | 0.451 | 0.061 | This work |

[a] Nearside and farside data used.

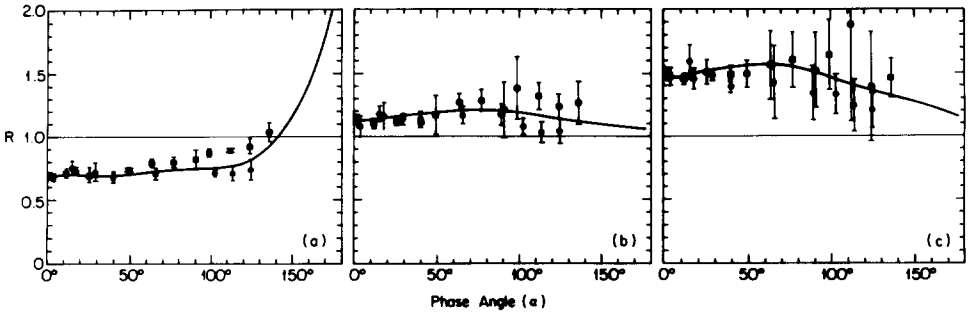



FIG. 4. $R$ vs $\alpha$ plots for bright (left), average (center), and dark (right) lunar terrains. Error bars are one standard deviation. Continuous curves represent $R$ predicted from disk-resolved best fit of Hapke's equation for hypothetical points such that $i = e = \alpha/2$.

earlier. Plots of reflectance ratio, $R$, against phase angle are presented in Fig. 4. Hapke's equation was fit to each of these data sets separately. In each case, the average disk-integrated values of Hapke's parameters were used as initial guesses. The approach of alternating between trials with fixed $\bar{\theta}$ and fixed $b$, used in the reduction of disk-integrated data, was also adopted. Numerical results are listed in the lower part of Table II. The curves in Fig. 4 represent the reflectance ratios predicted from the solutions for points oriented on the lunar surface such that $i = e = \alpha/2$ and are intended to demonstrate goodness of fit.

Values of Hapke's parameters derived from disk-resolved data for the three terrain groups are consistent with the solution obtained from the disk-integrated data. The value of each parameter derived from disk-integrated data lies between end-member dark and bright terrain values, and does not differ strongly from that derived for terrains of average albedo. Note that one should not expect "disk-integrated" values to correspond to numerical averages of the "disk-resolved" values. The Moon is geologically heterogeneous on scales much larger than typical regolith particles, and the disk-integrated photometric behavior is an areal combination of constituent terrain behaviors. We have not attempted to determine the appropriate relative weighting of "dark," "average," and "bright" terrains that would be required to adequately model the Moon's disk-averaged photometric behavior.

## V. DISCUSSION OF DISK-RESOLVED RESULTS

In the following discussion, we explore systematic variations in the photometric parameters found for the three different lunar terrains, and consider the physical interpretations of Hapke's parameters in relation to known properties of the lunar regolith. We also evaluate the geological significance of our results and the usefulness of Hapke's equation in modeling the photometric behavior of other celestial bodies.

Systematic differences in the values of each parameter between end-member dark and bright terrains can be identified in Table II. Not surprisingly, values for single scattering albedo ($w$) of the dominantly anorthositic average and bright terrains are consistently larger than the value for the basaltic dark terrains. In Fig. 5, we make use of the strong correspondence of single scattering albedo with terrain type to illustrate systematic trends in the remaining parameters.

Figure 5a shows that the $h$ parameter for dark terrains is significantly larger than for average and bright terrains; the latter two more closely match Hapke's value for the lunar farside. Larger values of $h$ cor-

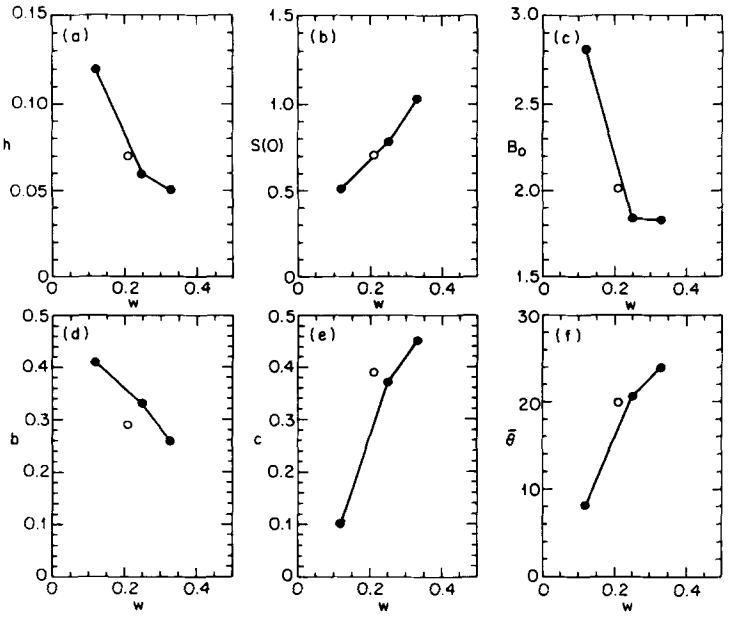



FIG. 5. Variations in photometric properties as a function of single scattering albedo. Solid dots represent results derived from disk-resolved observations; results derived from disk-integrated observations are shown by open circles. (a) $h$ vs $w$, (b) $S(0)$ vs $w$, (c) $B_0$ vs $w$, (d) $b$ vs $w$, (e) $c$ vs $w$, (f) $\bar{\theta}$ vs $w$. $w$ is the single scattering albedo. Other Hapke parameters are described in Table I.

respond to larger angular widths of the opposition effect. Hapke's (1986) equation provides an interpretation of $h$ in terms of regolith porosity ($1$-$F$), and particle-size distribution $n(r)$, where $r$ is particle radius. Which of these properties is more responsible for the systematic changes in $h$ with terrain type found above? Unfortunately, since different combinations of porosity and particle distributions can predict identical values of $h$, we can hope only to choose the simplest explanation from several equally valid possibilities.

In Hapke's regolith model, $h = -(\frac{3}{8})\ln(1-F)Y$, where $Y$ is a mathematical expression whose form depends on the particle-size distribution function, $n(r)$. Analytical expressions for $Y$ are given in Hapke's Table I for different particle-size distributions, including the simple power law, $n(r) = Kr^{-\beta}$, often used for characterizing soils. Here, $K$ is a constant which depends on the ratio, $(r_l/r_s)$, of the largest to smallest particle radii in the distribution and $\beta$ describes the shape of the particle distribution. Bhattacharya *et al.* (1975) show that $\beta = 4$ describes most lunar soil particles larger than about 15 $\mu$m and appears to be characteristic of comminution processes.

Following Hapke (1986), we begin by assuming a value of $(r_l/r_s) = 1000$ and compute implied porosities from our results (Table IV). The average, near-surface porosity of the lunar regolith (about 45%) is known to be remarkably uniform over the intercrater areas of the lunar surface (Houston *et al.* 1972, Mitchell *et al.* 1972, Carrier *et al.* 1972, 1973, 1974). Higher porosities (<70%) are observed primarily in crater ejecta and on steep topographic slopes (Houston *et al.* 1972) not typical of our sample areas. While our "disk-integrated" porosity (47%) is in good agreement with observed typical lunar values, the model porosities of dark and bright terrains are more inconsistent with observations and differ from each other by a factor of 2. Differences in surface porosity

TABLE IV

CONSISTENCY OF $h$ WITH MODEL REGOLITH PROPERTIES (ASSUMING PARTICLE-SIZE DISTRIBUTION $n(r) = K r^{-\beta}$)

| Material | $h$ | Porosity[a] (%) | $(r_l/r_s)^b$ | $\beta^c$ |
|---|---|---|---|---|
| Disk-integrated | 0.07 | 47 | 1,700 | 4.0 (1.4) |
| Dark terrains | 0.12 | 28 | 75 | 4.4 (1.0) |
| Average terrains | 0.06 | 53 | 5,700 | 3.9 (1.5) |
| Bright terrains | 0.05 | 58 | 32,000 | 3.7 (1.6) |

[a] Assuming $(r_l/r_s) = 1000$ and $\beta = 4.0$.
[b] Assuming porosity = 45% and $\beta = 4.0$.
[c] Assuming $(r_l/r_s) = 1000$ and porosity = 45%. Parenthetical numbers refer to second valid solution (see Fig. 6).

alone are unlikely to account for this difference. While $h$ depends on the rate of change of compaction with depth, the porosity of lunar soils probably does not change much over the topmost few millimeters to which optical remote sensing is most sensitive in the case of relatively opaque surfaces such as that of the Moon.

Another possible explanation for the differences in $h$ is a variation in the particle-size distributions. If we assume a porosity of 45% for all terrains and $\beta = 4$, Table IV shows the computed size ratio $(r_l/r_s)$. We see that size ratios differing by four orders of magnitude would be needed to account for variations in $h$, if terrains differed only by virtue of end-member particle size ratios. Observed particle-size distributions in returned lunar samples do, in fact, show $(r_l/r_s)$to be larger in highland regoliths than in mare (McKay *et al.* 1974), but not by several orders of magnitude.

As a final possibility, we can consider differences in the shapes of particle-size distributions, represented by variations in $\beta$. If we evaluate $\beta$ by assuming porosity of 45% and $(r_l/r_s) = 1000$ (solid curve in Fig. 6), then there are two solutions of $\beta$ for each value of h given in Table II (dashed lines in Fig. 6). One set of solutions centers about $\beta = 4$, which we know corresponds well to observed properties of lunar soils. Table IV presents both sets of solutions and demonstrates that only modest differences in $\beta$ are needed to account for the observed variations in $h$. If we accept solutions around $\beta = 4$ as most reasonable, then a slightly narrower particle-size distribution is implied for dark terrains than for brighter ones. Such a difference is consistent with observed grain-size distributions in lunar soil samples (cf. McKay *et al.* 1974). Thus, we conclude that the simplest physical explanation for the observed variations in $h$ is that particle-size distributions differ slightly for different lunar terrains, being narrower in the darker terrains.

Figures 5b and c show systematic terrain differences in the first-surface reflection term of the opposition amplitude, $S(0)$, and the total amplitude of the opposition surge, $B_0 = S(0)/w\{1 + b + c\}$, respectively. A lower limit on $S(0)$ for idealized euhedral lunar particles can be estimated from the Fresnel equation $S(0) = [(n - 1)^2 + k^2]/[(n + 1)^2 + k^2]$, where $n$ and $k$ are respectively the real and complex indices of refraction. For nonconducting minerals like

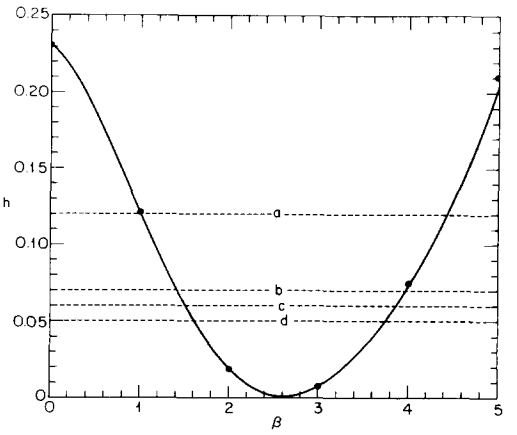



FIG. 6. Values of $h$ predicted for different particle-size distributions following the power law, $n(r) = Kr^{-\beta}$. Assumed porosity is 45%, and $r_l/r_s$ is 1000 (see text for discussion). Solid dots represent exact analytic solutions; curve is a least-squares fit of a sixth-order polynomial in $\beta$. Dashed lines correspond to values of $h$ listed in Table II for (a) dark terrains, (b) disk-integrated terrains, (c) average terrains, and (d) bright terrains. Values of $\beta$ corresponding to the intersections of dashed lines and solid curve are solutions for the corresponding terrains. The width of the particle-size distribution decreases with increasing $\beta$.

pyroxene, olivine, and anorthosite, $k \cong 0$ and $1.5 < n < 1.8$, giving $0.04 < S(0) < 0.10$. Using values of $w$, $b$, and $c$ from Table II would imply: $0.07 < B_0 < 0.55$. Table II shows that observed values of $S(0)$ and $B_0$ exceed these lower limits. In an idealized conducting soil of lunar ilmenite, $k \neq 0$, $n \sim k$, and $S(0)$ could be closer to unity. While ilmenite is a significant constituent of lunar soils, it is much less abundant than the dielectric minerals. Its role in elevating $S(0)$ could only be important if regolith particles were coated with ilmenitic iron to a significant degree (cf. Hapke *et al.* 1975). One would then expect to see larger values of $S(0)$ in the mare, where greater concentrations of ilmenite and other opaque minerals occur, contrary to the trend shown in Fig. 5.

An upper limit of $B_0 = 1$ was suggested by Hapke (1986) for opaque particles with smooth, regular surfaces. Figure 5c shows that all terrains exhibit $B_0 > 1$. According to Hapke (1986), $B_0$ can exceed unity (1) if particles are spherical and cause a glory-like backscatter, (2) if particles have highly crenulated shapes so that individual particles exhibit an opposition effect, or (3) if particles contain mechanical defects, such as cracks near particle surfaces. A glory-like backscatter from spherical particles like lunar glass beads may only contribute significantly to the opposition surges of local pyroclastic deposits where they occur in abundance. Generally, they do not occur in sufficient amounts to significantly contribute to the large opposition surges in our data. A more reasonable explanation is that lunar particles have highly irregular shapes and an abundance of mechanical defects due to micrometeoroid bombardment and disruption by impact shock. Agglutinates are particularly irregular in shape and are most abundant in lunar highland soils. This fact may partially account for the observation that $S(0)$ is highest for average and bright terrains. The constituent mineral crystals and glasses of highland soils are derived largely from anorthositic feldspars which are more transparent than the dark basaltic minerals that characterize mare soils. The greater ability of light to penetrate into highland regolith particles and reflect from the near-surface mechanical crystal defects could also contribute to the larger values of $S(0)$ for brighter terrains.

Curiously, while $S(0)$ is smallest for dark terrain materials, the total amplitude of the opposition surge, $B_0$, is largest. This result appears to be a consequence of the inverse proportionality of $B_0$ to single scattering albedo and that as particles become more opaque a larger fraction of the total amount of backscattered light comes from the first surface reflection, described by $S(0)$.

Differences in the particle phase function parameters, $b$ and $c$, are shown in Fig. 5d and e, respectively. The decrease in $b$ and increase in $c$ with increasing single scattering albedo is consistent with the expected tendency of higher albedo particles to be more forward scattering. The continuous decrease in $b$ corresponds to the decrease in the negative value (less backscattering) of the equivalent asymmetry factors: $-0.14$, $-0.11$, and $-0.09$ for dark, average, and bright terrains, respectively.

Perhaps the most remarkable variation with terrain type seen in our data is that of the average macroscopic roughness parameter, $\bar{\theta}$ (Table V). Values for average and

TABLE V

MACROSCOPIC ROUGHNESS FOR DIFFERENT PLANETARY OBJECTS

| Object | $\bar{\theta}$ | Source |
|---|---|---|
| Moon (disk-integrated) | 20° | This work |
| Moon: Dark terrain | 8° | This work |
| Average terrain | 21° | |
| Bright terrain | 24° | |
| Mercury (disk-integrated) | 20° | Hapke (1984) |
| | 21° | Veverka *et al.* (1987a) |
| Ganymede: Dark terrain | 25° | Helfenstein (1986) |
| Bright terrain | 3° | |
| Titania (disk-integrated) | 23° | Veverka *et al.* (1987b) |
| Titania: Dark terrain | 28° | Smith *et al.* (1986) |
| | 27° | Thomas *et al.* (1987) |
| Bright terrain | 21° | Thomas *et al.* (1987) |
| Io (disk-integrated) | 25° | Simonelli and Veverka (1987) |
| Europa | 23° | Buratti (1985) |
| Mimas | 30° | Buratti (1985) |

bright terrains are most like the "disk-integrated" value. The roughness of dark lunar terrains, however, appears to be less than half that of average and bright terrains. Our values of $\bar{\theta}$ are in good agreement with the earlier result of Wilson (1968, 1971) who found average slopes for mare ranging from 6° to 12° and slopes for highlands from 15° to 19° by application of Hapke's (1966) equation modified to model macroscopic roughness. These results are significant because prior to this investigation, few attempts to determine numerical measures of optical roughness for other planetary bodies yielded significant or easily interpretable variations. Pang *et al.* (1983) reported almost identical optical roughnesses for Phobos and Deimos, even though Deimos has a visibly smoother surface in Viking orbiter images extending in resolution down to scales of a few meters. The authors concluded that the optically detectable roughness of a body must represent small (subcentimeter) scales, a conclusion which implies that, under the influence of prolonged meteoroid bombardment, all airless planetary bodies might have the same optical roughness. Using computer topographic models of heavily cratered surfaces, Helfenstein (1986, 1987) verified that roughness is a cumulative property with decreasing scale and also showed that lunar-like crater-saturated surfaces predict values of $\bar{\theta}$ near 20°.

Values of $\bar{\theta}$ derived for different planetary bodies are listed for comparison in Table V, and the majority indeed have values of $20 < \bar{\theta} < 30°$. While most of these objects are heavily cratered, Io and Europa are not, at least at the kilometer-scale resolution of Voyager images. Because of the active resurfacing known to occur on Io, Simonelli and Veverka (1987) concluded that even in the absence of impact cratering, different geological processes could produce photometric roughness indistinguishable from that resulting from exposure to impacts.

Helfenstein (1986) found significantly different $\bar{\theta}$'s for terrrains on Ganymede (Table V) and concluded that, on that satellite, average bright terrains are smoother than average dark (ancient cratered) terrains. It is probably not coincidental that the photometrically smoothest terrains on both the Moon and Ganymede are those which are interpreted to have been resurfaced by some form of flood volcanism. Crater densities on these smoother terrains are generally lower and large scale topographic relief is generally less than on heavily cratered areas. These considerations suggest that, at least for the Moon and Ganymede, differences in roughness occur not only at visible (km) scales, but also at the comparatively small scales of optical detection. If roughness is indeed a cumulative property with decreasing scale, then mare should be smoother than highlands at scales intermediate between photometric (subcentimeter) and visual (kilometer). RMS slopes measured at radar wavelengths (13 cm) give decimeter scale slopes of about 4° for mare and 8° for highlands (Simpson and Tyler 1982). Thus, at radar wavelengths, the average highland RMS slope is twice that of the lunar mare. This ratio compares very well to the value of 2.5 found at optical scales from $\bar{\theta}$ (Table II).

The plots in Fig. 4 offer some additional information regarding roughness variations on the lunar surface. Regardless of terrain type, all points falling below the best-fit curves at high phase angles correspond to samples obtained near the eastern limb of the Moon, a region primarily covered by highlands materials. All points falling on or above the curves at high phase angles were obtained near the western limb, an area dominated by mare. The simplest explanation for this trend is that the east-limb terrains are systematically rougher than their west-limb counterparts.

The circles plotted in Figs. 5a–f show that the parameters derived from disk-integrated data agree well with those derived from disk-resolved observations of average albedo terrains. This correspon-

dence is nontrivial because disk-integrated phase curves and disk-resolved reflectance measurements are different physical quantities. The good mutual agreement of results obtained from both data sets is an independent check on the validity of our solutions. It will be recalled that Shorthill *et al.* (1969) classified these terrains solely on the basis of normal albedos. Our results show that there is a strong correspondence of other important lunar photometric parameters with albedo.

## VI. CONCLUSIONS

We have verified that, in the case of the Moon, physically and geologically interpretable parameters can be derived from disk-integrated and disk-resolved photometric measurements using Hapke's equation. The single scattering albedos of average and bright lunar terrains are systematically larger than those of lunar mare. Average particles in dark terrain regoliths are consistently more strongly backscattering than those in average and bright lunar terrains. The greater angular width of the opposition surge for dark terrains is most easily explained by modest differences in regolith particle-size distributions which might accompany the normal regolith maturation process. The trend is such that darker terrains should have narrower particle-size distributions. The total amplitude of the opposition surge for dark terrains is larger than for average and bright terrains. This result appears to be a consequence of the fact that in opaque particles, a larger fraction of singly scattered light at zero phase comes from first-surface reflection. The average subcentimeter macroscopic roughness of dark terrains is significantly lower than that of average and bright terrains; the magnitude of this difference is consistent with that inferred from radar measurements at decimeter scales.

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